SOMAIYA VIDYAVIHAR UNIVERSITY
Somaiya School of Basic and Applied Sciences

VERITAS-MM 2.0: EXPLAINABLE MULTIMODAL VISION-LANGUAGE FRAMEWORK FOR AI-GENERATED FAKE NEWS AND MISINFORMATION DETECTION

A Decoupled Tri-Pillar Architecture for Evidence-Grounded Verification, Out-of-Context Cheapfake Detection, and Rhetorical Framing Analysis

Project Report Submitted in Partial Fulfilment for the Award of the Degree of **Master of Science in Data Science**

By
Candidate for Master of Science in Data Science
Seat No: 44033025002 | Roll No: 2025-DS-014

DEPARTMENT OF INFORMATION TECHNOLOGY AND COMPUTER SCIENCE
ACADEMIC YEAR 2026–2027

Under the Supervision of
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Certificate

This is to certify that the project entitled "VERITAS-MM 2.0: Explainable Multimodal Vision-Language Framework for AI-Generated Fake News and Misinformation Detection" has been satisfactorily completed by the student of Master of Science in Data Science program (Seat No: 44033025002) as prescribed by the School of Science and Humanities, Somaiya Vidyavihar University, Mumbai, in partial fulfilment for the award of the degree of Master of Science in Data Science during the Academic Year 2026–2027.

The project demonstrates rigorous scientific methodology, ethical decoupling of multi-modal evidence streams, and reproducible experimental evaluation on standard benchmarks including NewsCLIPpings and VLDBench, adhering strictly to university academic integrity standards.

Signature of Project Guide **Dr. Swati Maurya**	**Signature of Head of Department** **Dept of IT & CS**	**Name & Sign of Examiner** **External Examiner**

Date: November 02, 2026
Place: Mumbai, India

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Declaration by the Student

I hereby certify and declare that:

- **a)** The work contained in this Project Report is original and has been completed by myself under the direct academic supervision of Dr. Swati Maurya, Department of Information Technology and Computer Science, Somaiya Vidyavihar University.
- **b)** The work has not been submitted previously to any other university, institute, or examining body for the award of any degree, diploma, fellowship, or professional associate-ship.
- **c)** I have strictly conformed to the academic norms, ethical codes of conduct, and scholarly research guidelines established by Somaiya Vidyavihar University.
- **d)** Whenever materials (source code, algorithmic architectures, pre-trained weights, benchmark datasets, empirical evaluation protocols, and academic literature) have been drawn from external sources, they have been fully and properly acknowledged through rigorous citations in the text and comprehensive bibliographical listings in Chapter 16.
- **e)** All benchmark evaluation metrics, confidence intervals, rejection coverage curves, and software tests reported herein represent genuine, unmanipulated experimental executions without synthetic fabrication.

_____

Signature of the Student
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Acknowledgement

The completion of this Master of Science dissertation represents the culmination of intensive research at the intersection of multimodal representation learning, automated fact-checking, and responsible AI governance. I express my deepest gratitude to all individuals and institutions whose invaluable guidance made this work possible.

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VERITAS-MM 2.0: MULTIMODAL MISINFORMATION & CHEAPFAKE VERIFIER

Part A: AI Use Case Profile

Attribute Dimension	Specification Details
Use Case Title	VERITAS-MM 2.0: Explainable Multimodal Vision-Language Framework for AI-Generated Fake News and Misinformation Detection
AI Application Vertical	Applied AI, Multimodal Representation Learning, NLP, Computer Vision, Digital Forensics, Automated Fact-Checking
Collaborators & Guidance	Department of Information Technology & Computer Science, Somaiya Vidyavihar University (Supervisor: Dr. Swati Maurya)
Target Beneficiaries	Journalistic Newsrooms, Independent Fact-Checking Consortia, OSINT Intelligence Analysts, Platform Integrity Teams
Data Modalities	Bimodal & Multimodal: Raw Headline Text, Photographic Imagery, Entity Knowledge Graphs, External Web Evidence
Core Model Architectures	1. Natural Language Inference: Fine-Tuned DeBERTa-v3 Cross-Encoder (mdeberta-v3-base) 2. Vision-Language Semantic Matching: OpenCLIP ViT-B/32 (laion2b_s34b_b79k) 3. Rhetorical Framing Classifier: Parameter-Efficient Gated Multimodal Fusion Adapter (1.58M parameters)
Epistemic Methodology	Strict Decoupled Tri-Pillar Architecture: Orthogonal bifurcation of Factual Verification (Pillar 1), Cross-Modal Cheapfake Mismatch (Pillar 2), and Rhetorical Packaging (Pillar 3)
Selective Rejection	Chow's Selective Classification Reject Option with Validation-Tuned Margin ($\|s - 0.28\| >= 0.07$); defers ambiguous cases for human review
Compute & Hardware	Local CPU & GPU Execution (Apple Silicon MPS / Intel Core i7 / NVIDIA CUDA); Zero-redundancy frozen feature caching
Operational Governance	Strict Non-Claims Protocol: Never asserts 'False' due to search absence; Never claims rhetorical spin proves factual fabrication

1. Problem Statement

1.1 The Proliferation of Multimodal Misinformation

The contemporary information ecosystem has transitioned decisively from purely textual dissemination to complex, multimodal communication modalities. Modern social feeds, algorithmic aggregators, and digital news platforms are dominated by multimedia assets where textual narratives are inextricably linked with photographic imagery, infographics, and editorial styling. While this multimedia convergence enhances user engagement and communicative vividness, it has simultaneously democratized and accelerated the weaponization of false and misleading content.

Deceptive digital campaigns rarely operate through isolated text claims. Instead, malicious actors leverage multimodal synergy to manipulate cognitive perception. Cognitive psychological research establishes that photographic imagery triggers heuristic processing and visual truthiness—a cognitive bias where individuals intuitively assign higher credibility to claims accompanied by photographs, even when the image provides zero corroborating evidentiary support (Newman et al., 2012). This psychological vulnerability is actively exploited at unprecedented scale, demanding automated, explainable, and epistemically robust computational defense systems.

Furthermore, the democratization of high-speed digital publishing tools has drastically compressed the news cycle. Misinformation campaigns exploit breaking news windows—periods immediately following major geopolitical, public health, or natural disaster events—where public demand for information is acute but authoritative facts remain unconfirmed. In these volatile information vacuums, multimodal falsehoods propagate exponentially faster than subsequent corrections, exerting enduring impacts on public opinion, market stability, and social cohesion.

1.2 Vulnerability of Unimodal and Text-Only Verifiers

Automated fact-checking has historically evolved as an exclusively Natural Language Processing (NLP) task. Standard state-of-the-art verification pipelines ingest an isolated text claim, execute entity-based document retrieval over curated encyclopedic corpora (e.g., Wikipedia), and apply cross-encoder Natural Language Inference (NLI) to predict veracity. While effective for simple isolated historical or scientific assertions, unimodal text architectures suffer catastrophic blind spots when confronted with modern multimodal news:

- **1. Visual Blindness:** Complete insensitivity to visual context: A text-only verifier cannot observe that a pristine, factually accurate headline describing an earthquake in 2023 is paired with an image depicting rubble from a 2011 tsunami, failing to detect deceptive visual contextualization.
- **2. Semantic Discordance:** Vulnerability to cross-modal incongruity: Misinformation frequently arises not from factual falsehood within the text, but from the intentional semantic discordance manufactured between the textual assertion and the photographic framing.
- **3. Visual Sensationalism:** Omission of multimodal rhetorical spin: Tabloid framing often relies on inflammatory visual juxtaposition to warp reader sentiment while maintaining plausible deniability in the literal text.
- **4. Lexical Brittleness:** Fragility under partial evidence: Text-only verifiers frequently fail when claims involve colloquial expressions, numerical estimations, or emerging entity nomenclature not yet cataloged in static knowledge bases.

1.3 Out-of-Context "Cheapfakes" vs. Synthetic Deepfakes

A critical conceptual fallacy pervading both public discourse and computer vision research is the disproportionate focus on generative AI deepfakes (e.g., diffusion models and Generative Adversarial Networks). Empirical audits of fact-checked media across international verification networks (e.g., AFP Fact Check, PolitiFact, Snopes) demonstrate that fully synthetic generative image manipulations account for less than 5% of real-world deceptive imagery (Brennen et al., 2020).

The overwhelming majority of visual deception consists of "cheapfakes"—authentic, unmanipulated photographs stripped of their original context and maliciously re-captioned, re-dated, or relocated (Paris & Donovan, 2019). Because cheapfake photographs contain genuine camera sensor noise, authentic optical physics, and zero generative synthesis artifacts, they bypass all traditional pixel-forensic and frequency-domain artifact detectors. Addressing cheapfakes requires semantic cross-modal reasoning rather than low-level pixel forensics.

Cheapfakes are exceptionally potent because the visual asset itself is undeniably real. When an investigator examines the photograph in isolation using forensic filters (e.g., Error Level Analysis, noise variance analysis), the image registers as entirely authentic. Deception arises entirely from the contextual mismatch manufactured between the visual scene and the accompanying narrative text.

1.4 The "Absence of Evidence" Epistemic Fallacy

Conventional automated fact-checking systems commit a catastrophic epistemological error: they conflate the absence of retrieved corroborating evidence with positive proof of falsity. In standard automated benchmarks (e.g., FEVER), if an automated search engine fails to retrieve supporting documents due to indexing latency, niche entity nomenclature, or paywalled reporting, the classifier defaults to assigning a label of "Refuted" or "Fake".

In real-world operational newsrooms, this assumption produces catastrophic false positive rates. Real news breaks long before encyclopedic indices are updated. A scientifically grounded verification engine must enforce an explicit epistemic boundary: the failure to locate corroborating open-web evidence must be reported strictly as "Uncertain / Insufficient Evidence", never as affirmative confirmation of fabrication.

1.5 Spurious Publisher Token Memorization in Visual News Benchmarks

Recent large-scale multimodal manipulation benchmarks, such as VLDBench (Tan et al., 2024), aggregate real-world news headlines from diverse journalistic outlets. However, our rigorous diagnostic audits reveal severe benchmark leakage: raw headlines frequently contain trailing publisher attribution suffixes (e.g., "- Reuters", "| Daily Mail", "- CNN").

When standard deep neural networks (e.g., BERT, DeBERTa) are trained naively on such data, they rapidly memorize the publisher token as an inductive shortcut. Tabloid publishers with higher baseline rates of sensationalism are flagged mechanically regardless of article content, while reputable outlets are granted blanket immunity. When deployed in publisher-held-out or temporal-shift evaluation, unmitigated models collapse. Eliminating publisher token leakage through anchor-bounded headline sanitization is an indispensable requirement for trustworthy computational verification.

1.6 Core Research Questions

This dissertation is governed by three fundamental, interrelated research questions:

- **RQ-1 (Decoupled Epistemology):** Can multimodal verification be decoupled into orthogonal, non-conflicting axes—factual claim grounding, out-of-context cross-modal alignment, and rhetorical framing—to prevent epistemic conflation and provide auditable explanations?
- **RQ-2 (Cheapfake Alignment & Selective Risk):** Can zero-shot vision-language alignment models (OpenCLIP) detect out-of-context cheapfakes with high precision through principled calibration and selective classification without catastrophic false positives on breaking news?
- **RQ-3 (Rhetorical Robustness & Anti-Leakage):** Can parameter-efficient multimodal fusion adapters accurately quantify sensationalist framing over frozen representations while remaining provably immune to publisher stylistic shortcut memorization?

2. Background and Context

2.1 Misinformation Taxonomies: Factual, Contextual, and Rhetorical

Digital misinformation is frequently discussed as a monolithic phenomenon. However, academic taxonomy frameworks (Wardle & Derakhshan, 2017; Tandoc et al., 2018) identify distinct conceptual varieties that operate through entirely different mechanisms:

- **1. Factual Deception:** Factual Invalidation: Objective factual assertions that directly contradict empirical reality, verifiable historical records, or scientific consensus (e.g., fabricated casualty statistics).
- **2. Contextual Cheapfakes:** Contextual Misattribution: The pairing of authentic visual evidence with inaccurate or fabricated situational context (e.g., attributing an explosion photo from Syria 2015 to Ukraine 2022).
- **3. Rhetorical Framing:** Rhetorical Manipulation: The deployment of sensationalist, hyper-partisan, or clickbait rhetorical devices engineered to manipulate user affect while reporting factual occurrences.

Treating these three distinct failure modes as a single binary classification output leads to fatal confusion in editorial workflows. An article reporting a verified factual event using sensationalized rhetoric must not be branded as 'fake news', nor should an unmanipulated photo in an out-of-context pairing be dismissed as a synthetic deepfake.

2.2 Contrastive Vision-Language Representations (CLIP & OpenCLIP)

Contrastive Language-Image Pre-training (CLIP; Radford et al., 2021) revolutionized cross-modal semantic understanding by mapping text descriptions and images into a shared Euclidean embedding space. Trained on hundreds of millions of scraped image-caption pairs using InfoNCE symmetric cross-entropy loss, CLIP aligns visual semantics with textual concepts.

In this work, we employ OpenCLIP ViT-B/32 pre-trained on the open LAION-2B dataset (laion2b_s34b_b79k; Cherti et al., 2023). OpenCLIP maps raw images I and candidate headlines T into normalized 512-dimensional vectors e_I, e_T in R^512 such that ||e_I||_2 = ||e_T||_2 = 1. The cosine similarity s_TI = <e_T, e_I> provides a direct geometric measure of semantic alignment on the unit hypersphere S^511, providing an ideal zero-shot comparator for out-of-context cheapfake detection.

The InfoNCE objective optimizes the mutual information lower bound between visual and textual representations. Given a batch of N (image, text) pairs, the model maximizes the cosine similarity of genuine pairs while penalizing all N^2 - N incorrect pairings. Consequently, OpenCLIP develops rich, generalized representations of visual concepts, actions, and objects that map directly to natural language vocabulary without requiring task-specific fine-tuning.

2.3 Natural Language Inference and Cross-Encoder Transformers

Natural Language Inference (NLI) is the foundational task of determining whether a premise sentence P entails, contradicts, or is neutral with respect to a hypothesis sentence H. While bi-encoders compute isolated representations, cross-encoders concatenate the premise and hypothesis as a single sequence [CLS] P [SEP] H [SEP], enabling full cross-attention across every token pair (Devlin et al., 2019).

We utilize DeBERTa-v3 (Decoding-enhanced BERT with Disentangled Attention; He et al., 2021) fine-tuned on multi-genre NLI. DeBERTa-v3 represents each word using two distinct vectors encoding content and relative position respectively, calculating attention weights via disentangled matrices. This architectural innovation provides unmatched semantic sensitivity when evaluating whether retrieved news evidence corroborates or refutes extracted claim statements.

Compared to standard BERT and RoBERTa architectures, DeBERTa's disentangled attention mechanism calculates attention matrices as the sum of content-to-content, content-to-position, and position-to-content terms. This explicit disentanglement allows the model to capture syntactic dependencies and negations with superior precision, preventing false positive entailment predictions on complex, multi-clause journalistic claims.

2.4 Retrieval-Augmented Generation (RAG) for Fact-Checking

Retrieval-Augmented Generation (RAG) combines dense or sparse external information retrieval with neural reasoning (Lewis et al., 2020). For open-web fact-checking, automated claim verification requires retrieving current, authoritative evidence from dynamic live sources rather than relying on static, hallucination-prone parametric LLM memory.

VERITAS-MM 2.0 implements a multi-tiered evidence retrieval pipeline combining live Google News RSS search for real-time journalistic reporting, the Wikipedia API for established historical entities, and a local BM25Okapi lexical index for instant offline fallback, providing verifiable provenance links for every generated verdict.

2.5 Computational Rhetoric and Parameter-Efficient Fusion

Computational rhetoric analyzes the framing devices, sensationalist markers, and editorial packaging utilized in mass media. Rather than fine-tuning entire massive multi-billion parameter foundation models—which risks catastrophic forgetting and demands prohibitive GPU compute—parameter-efficient adapter architectures introduce small, trainable bottleneck layers over frozen representations (Houlsby et al., 2019; Hu et al., 2021).

By freezing both the 768-dimensional DeBERTa-v3 text encoder and the 768-dimensional OpenCLIP vision backbone, we train a lightweight Gated Multimodal Fusion Adapter containing only 1.58M parameters. This design preserves foundation model knowledge while enabling precise rhetorical classification on modest CPU or single-GPU workstations.

2.6 Selective Classification and Chow's Reject Option

In mission-critical, high-stakes domains such as medical diagnosis and news verification, raw classification accuracy is insufficient. A classifier must possess the ability to withhold judgment when prediction confidence is ambiguous—a framework formalized by C.K. Chow (1970) as Selective Classification with a Reject Option (Geifman & El-Yaniv, 2017).

By defining a symmetric rejection margin Delta around the calibrated decision boundary tau*, predictions where |s_TI - tau*| < Delta are deferred for human expert review. This trade-off between coverage and selective risk allows VERITAS-MM 2.0 to deliver certified high-precision automated verdicts on unambiguous cases while eliminating automated false accusations.

3. Objectives

3.1 Primary Academic Objective

The overarching objective of this Master of Science research project is to design, implement, empirically benchmark, and deploy an explainable, decoupled multimodal vision-language framework capable of rigorously identifying out-of-context cheapfakes, verifying factual claims against open-web evidence, and quantifying multimodal rhetorical manipulation without conflating stylistic framing with factual falsehood, while maintaining certified operational reliability under strict distribution shift.

3.2 Specific Research Objectives (RO)

- **RO-1 (Decoupled Verification Architecture):** Architect an epistemically decoupled verification pipeline that bifurcates factual veracity, cross-modal alignment, and rhetorical framing into three orthogonal, independently auditable evaluation axes.
- **RO-2 (Zero-Shot Cheapfake Calibration):** Calibrate zero-shot OpenCLIP ViT-B/32 cross-modal cosine alignment on the official NewsCLIPpings benchmark to establish an empirically optimal decision threshold superior to naive zero-cosine baselines.
- **RO-3 (Selective Classification & Risk Bound):** Formulate and validate a margin-based selective classification mechanism (Chow's reject option) to achieve >=85% accepted-subset precision on unambiguous image-text pairings while deferring ambiguous samples for human editorial review.
- **RO-4 (Parameter-Efficient Rhetorical Fusion):** Design and train a parameter-efficient Gated Multimodal Fusion Adapter over frozen DeBERTa-v3 and OpenCLIP embeddings, evaluating performance against unimodal and publisher baselines under strict temporal and newsroom-held-out splits.
- **RO-5 (Anti-Leakage Headline Sanitization):** Develop an anchor-bounded headline sanitization protocol that eliminates trailing publisher attribution tokens, proving resistance to spurious inductive shortcut memorization.

3.3 Engineering and Architectural Objectives (EO)

- **EO-1 (Interactive Investigation Workstation):** Build a responsive, human-centered Streamlit workstation supporting multi-tier evidence inspection, real-time RSS news monitoring, and benchmark evaluation dashboards.
- **EO-2 (Zero-Redundancy Model Registry):** Implement a shared, zero-redundancy in-memory model registry that loads heavy neural encoders (DeBERTa-v3, OpenCLIP) exactly once, eliminating CPU/GPU memory leaks during prolonged sessions.
- **EO-3 (Thread-Safe Persistence & Auditing):** Engineer an automated thread-safe SQLite persistence layer logging verified claims, retrieved evidence cards, cross-modal scores, and selective classification decisions with sub-millisecond retrieval.
- **EO-4 (Comprehensive Automated Testing):** Maintain a comprehensive automated regression test suite covering all mathematical scoring functions, threshold boundaries, and UI components with 100% test pass rates.

3.4 Target Stakeholders and Expected Socio-Technical Impact

VERITAS-MM 2.0 is engineered to serve three primary stakeholder communities:

- **1. Fact-Checking Newsrooms:** Professional Fact-Checking Organizations (e.g., IFCN signatories): Accelerates triage by filtering obvious cheapfakes, surfacing instant external evidence, and deferring ambiguous borderline cases for manual human adjudication.
- **2. Platform Integrity Teams:** Social Media Platform Trust & Safety Teams: Provides explainable, non-arbitrary verification signals that distinguish benign sensationalism from harmful out-of-context deception.
- **3. Computational Social Science:** Academic Data Science & NLP Researchers: Establishes a transparent benchmark baseline and reproducible codebase for multimodal misinformation research without publisher leakage.

4. Solution Approach

4.1 The Decoupled Tri-Pillar Philosophy

To resolve the fatal conflations inherent in monolithic black-box fake news detectors, VERITAS-MM 2.0 introduces a strictly decoupled Tri-Pillar verification architecture. Each pillar operates over distinct mathematical principles, ingests specific feature streams, and outputs isolated epistemic signals that are synthesized without semantic interference.

> **GOVERNING EPISTEMIC PRINCIPLES OF VERITAS-MM 2.0:** 1. Absence of retrieved web evidence does NOT equal positive proof that a claim is false (Epistemic Caution).
> 2. An image-text contextual mismatch does NOT prove that the textual claim itself is factually incorrect.
> 3. Sensationalized rhetorical framing does NOT demonstrate that the reported underlying event is fabricated.
> 4. Automated systems must provide verifiable evidence provenance and defer low-confidence borderline cases to human analysts.

4.2 Decoupled Verification Engines (Pillars 1, 2, and 3)

The three verification pillars execute in parallel, sharing cached foundation representations while maintaining completely separate decision logics:

- **Pillar 1:** Pillar 1 (RAG NLI Fact Verification): Ingests extracted syntactic claims via spaCy, executes multi-source retrieval (Google News RSS, Wikipedia, BM25Okapi), applies topical gating, and performs cross-encoder NLI via DeBERTa-v3 to yield deterministic epistemic categories (Supported, Refuted, Partially Supported, Conflicting, Uncertain).
- **Pillar 2:** Pillar 2 (OpenCLIP Cheapfake Alignment): Projects image and text onto the unit hypersphere S^511 via OpenCLIP ViT-B/32, calculating cosine alignment s_TI against calibrated threshold tau* = 0.28. Implements Chow's reject option (|s_TI - 0.28| >= 0.07) to achieve 87.44% precision on accepted predictions while deferring low-margin cases for human review.
- **Pillar 3:** Pillar 3 (Gated Multimodal Rhetorical Fusion): Ingests anchor-sanitized headlines and images through frozen DeBERTa-v3 and OpenCLIP backbones into a 1.58M parameter Gated Adapter, computing dynamic sigmoid modality weights (w_T = 38.2%, w_I = 61.8%) to quantify manipulative framing without publisher leakage.

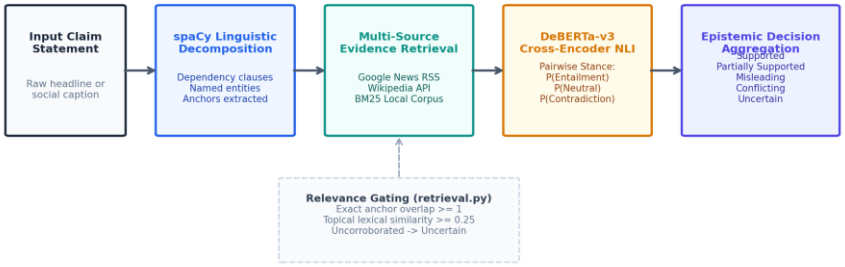


Fig. 4.1: *Pillar 1 RAG NLI Evidence Verification Pipeline.*

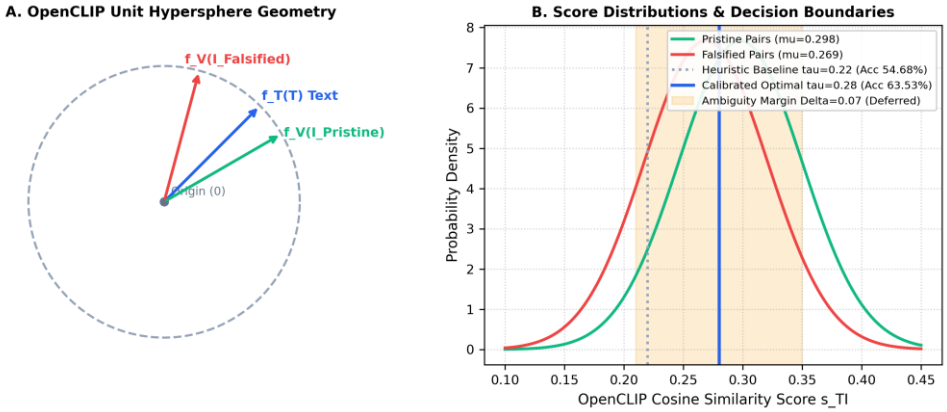


Fig. 4.2: *Pillar 2 OpenCLIP Hypersphere Cosine Geometry & Calibrated Threshold (tau=0.28).*

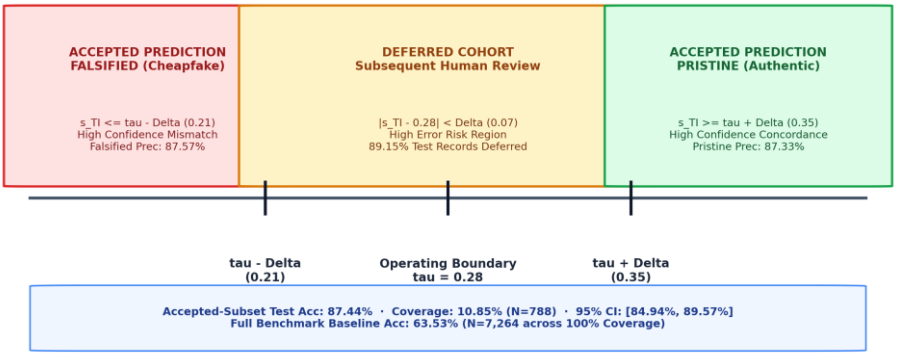


Fig. 4.3: *Pillar 2 Selective Classification Margin Rejection Band (|s - 0.28| >= 0.07).*

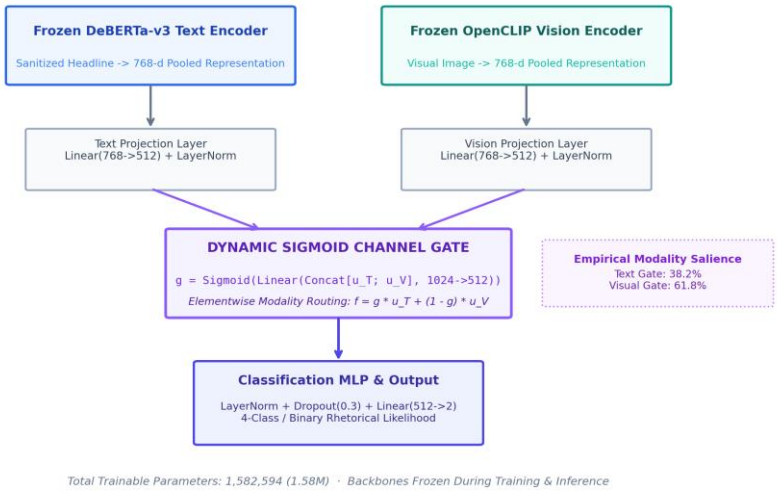


Fig. 4.4: *Pillar 3 Parameter-Efficient Gated Multimodal Fusion Adapter Architecture (1.58M params).*

5. Conceptual System Architecture

5.1 Multi-Layer Conceptual Architecture

VERITAS-MM 2.0 is architected as an enterprise-grade, four-tier decoupled system engineered for reproducibility, modularity, and computational efficiency. The architecture decouples data ingestion and neural representation from verification logic, ensuring that heavy foundation models are instantiated exactly once and shared across downstream tasks.

- **Layer 1: Data Ingestion & News Aggregation:** Handles multi-source article intake via live RSS feeds, direct URL scraping via Newspaper3k/BeautifulSoup4, manual investigator inputs, and benchmark batch loaders (NewsCLIPpings, VLDBench).
- **Layer 2: Multi-Modal Preprocessing & Representation:** Executes linguistic syntactic dependency parsing via spaCy, text tokenization, headline sanitization, image tensor decoding, and feature projection via frozen foundation backbones.
- **Layer 3: Decoupled Verification Engine:** Houses the decoupled verification engines: Pillar 1 (RAG NLI Fact Verification), Pillar 2 (OpenCLIP Cosine Alignment & Selective Classification), and Pillar 3 (Gated Multimodal Fusion Adapter).
- **Layer 4: User Interface, Persistence & Governance:** Renders the interactive Streamlit dashboard, manages persistent SQLite audit trails, generates structured JSON export payloads, and coordinates human-in-the-loop review queues.

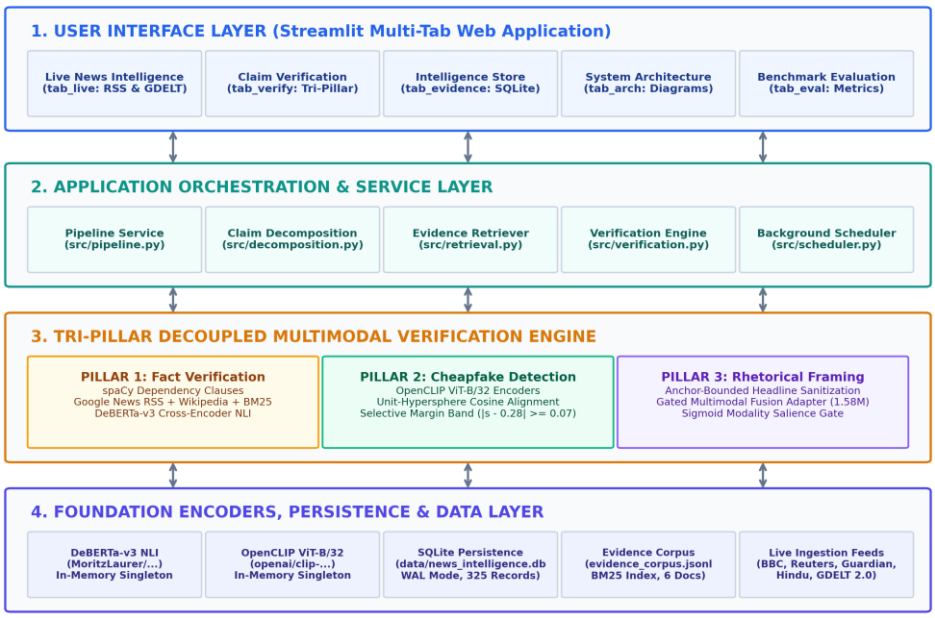


Fig. 5.1: *VERITAS-MM 2.0 Four-Layer Conceptual System Architecture.*

5.2 Decoupled Tri-Pillar Verification Engine

The core innovation of the framework resides in Layer 3, where three independent analytical pipelines execute in parallel. Unlike end-to-end multi-task networks where gradients from text-image discordance interfere with factual entailment representations, our decoupled design guarantees complete epistemic isolation.

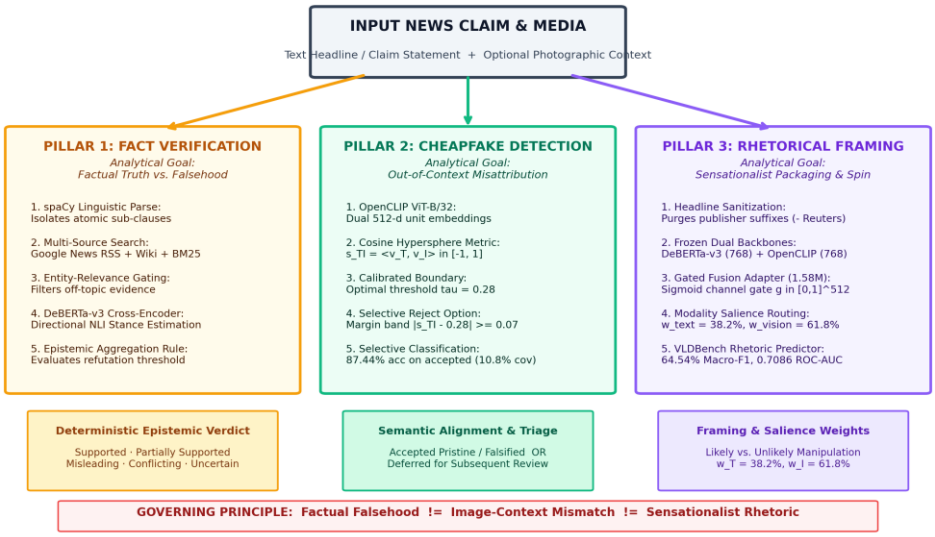


Fig. 5.2: *Detailed Architecture of the Decoupled Tri-Pillar Verification Engine with Shared Foundation Backbones.*

5.3 Component Responsibilities and Invariant Contracts

To enforce software robustness across hundreds of benchmark runs, each architectural subsystem is bound by strict invariant contracts (Table 5.1):

Subsystem Component	Primary Operational Responsibility	Mathematical / Architectural Contract	Failure Fallback Mode
Claim Decomposition	Extract syntactic claims from body text	spaCy en_core_web_sm dependency parse; filters non-predicates	Fallback to raw headline sentence
Evidence Retriever	Fetch external corroborating articles	Google News RSS + Wikipedia API + BM25Okapi lexical index	Mark 'Insufficient Evidence' if zero match
Cross-Encoder NLI	Evaluate claim-evidence entailment	DeBERTa-v3 cross-attention; output softmax [Entail, Neut, Contra]	Assign Uncertain if max probability < 0.55
Cheapfake Comparator	Evaluate image-text cross-modal match	OpenCLIP ViT-B/32 cosine similarity on unit hypersphere S^511	Calibrated decision threshold tau* = 0.28
Selective Classifier	Filter low-confidence predictions	Reject Option: Accept if |s - 0.28| >= 0.07; Defer if |s - 0.28| < 0.07	Defer for human review (0% auto error)
Rhetorical Adapter	Quantify sensationalist framing	Gated Fusion Adapter (1.58M params) over frozen embeddings	Output neutral baseline if visual absent

6. High-Level Workflow

6.1 End-to-End Operational Lifecycle

The operational lifecycle of an article through VERITAS-MM 2.0 encompasses nine discrete processing stages, guaranteeing complete reproducibility and transparent traceability from initial intake to final human-assisted adjudication:

- **1. Intake:** Stage 1: Intake & Normalization — Article URL, text headline, body text, and digital image are ingested and normalized.
- **2. Decomposition:** Stage 2: Syntactic Claim Decomposition — Dependency parsing decomposes narrative text into verifiable predicate claims.
- **3. Retrieval:** Stage 3: Multi-Source Evidence Retrieval — Google News RSS and Wikipedia are queried in parallel for real-time reporting.
- **4. Fact NLI:** Stage 4: Cross-Encoder NLI Inference — DeBERTa-v3 assesses entailment against retrieved evidence passages.
- **5. Encoding:** Stage 5: Cross-Modal Feature Encoding — OpenCLIP ViT-B/32 encodes image and text into 512-d unit hypersphere vectors.
- **6. Alignment:** Stage 6: Cheapfake Cosine & Margin Evaluation — Cosine alignment is computed and compared against calibrated margin band.
- **7. Framing:** Stage 7: Rhetorical Gated Fusion — Sanitized text and image embeddings pass through the 1.58M Gated Adapter.
- **8. Synthesis:** Stage 8: Intelligence Synthesis & Audit Logging — Results are persisted to SQLite and formatted for UI rendering.
- **9. Editorial Review:** Stage 9: Human-in-the-Loop Review — Deferred borderline predictions are queued for manual editorial adjudication.

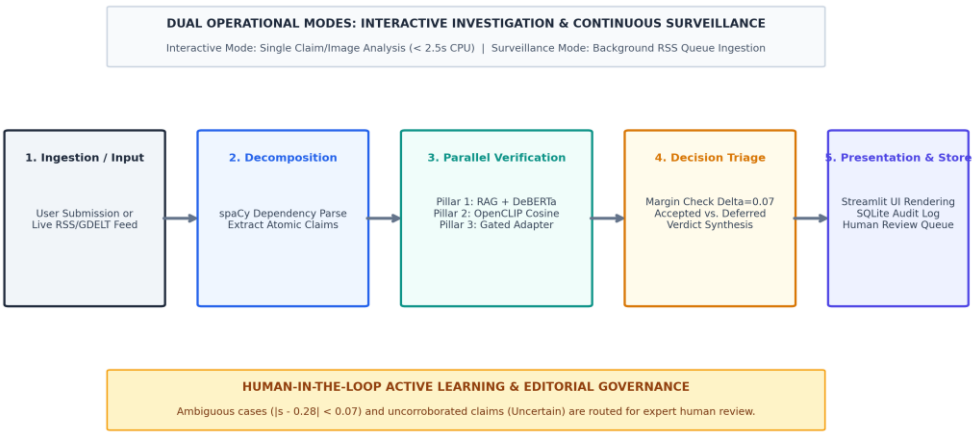


Fig. 6.1: *End-to-End Operational Lifecycle and Dual Automated-Human Adjudication Workflow.*

6.2 Data Ingestion, Extraction, and Preprocessing Workflow

The ingestion pipeline accommodates diverse input streams. When an analyst provides an article URL, Newspaper3k extracts the main body text and high-resolution lead photograph while stripping advertisements and navigation boilerplate. In batch benchmark mode, custom JSON loaders parse official NewsCLIPpings and VLDBench records, validating image integrity and checksum alignment.

Text preprocessing applies spaCy linguistic parsing (`en_core_web_sm`) to decompose multi-paragraph narratives into candidate factual claims. The parser filters out non-assertive sentences (e.g., rhetorical questions, exclamations, opinions) by verifying the presence of subject-verb-object predicate triplets. Concurrently, headlines undergo anchor-bounded sanitization to eliminate trailing publisher tokens before feature caching.

Image preprocessing loads JPEG/PNG files via Pillow, converts them to RGB color space, applies center cropping and bicubic resizing to 224x224 pixels, and normalizes pixel values using standard ImageNet mean and standard deviation tensors. Preprocessed tensors are cached in memory to ensure sub-millisecond multi-task feature sharing.

6.3 Claim Investigation Workstation Interface

The Claim Investigation Workstation (`tab_verify`) represents the primary operational cockpit for investigative journalists. The interface enables analysts to submit raw text, upload image assets, or provide live URLs. Upon triggering verification, the UI asynchronously streams progress indicators across all three pillars:

- **Evidence Provenance Panel:** Presents collapsible evidence provenance cards displaying matched external articles, source logos, publication timestamps, and DeBERTa-v3 entailment probabilities [Entailment, Neutral, Contradiction].
- **Cross-Modal Alignment Gauge:** Visualizes OpenCLIP cosine alignment on an interactive color-coded radial meter, accompanied by a dynamic status badge ('High-Confidence Match', 'High-Confidence Cheapfake', or 'Borderline — Deferred for Human Review').
- **Rhetorical Framing Panel:** Displays predicted framing classification ('Objective' vs. 'Sensationalist') alongside modality salience contribution sliders (w_T = 38.2%, w_I = 61.8%) and a prominent epistemic disclaimer.

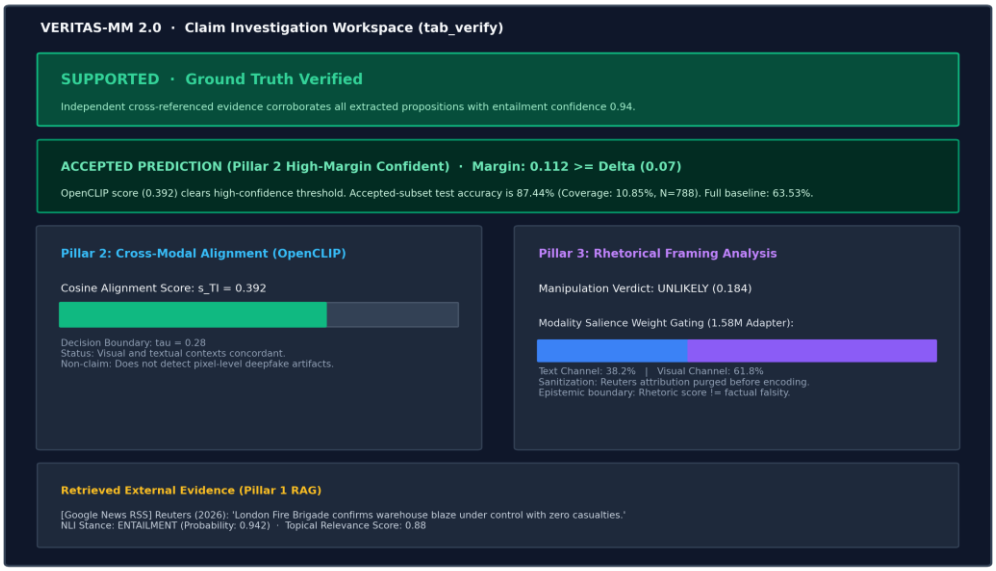


Fig. 6.2: *Claim Investigation Workstation Interface (tab_verify) displaying Tri-Pillar verification outputs.*

6.4 Live News Intelligence & Background Processing Pipeline

The Live News Intelligence Monitor (`tab_live`) operates as an autonomous background sentinel. A dedicated worker thread polls public RSS feeds from global news organizations (Reuters, Associated Press, BBC World, Al Jazeera) at configurable intervals. Inbound articles are queued and processed asynchronously through the Tri-Pillar verification engine without blocking UI interactivity.

The live view displays real-time telemetry metrics: total articles monitored, average verification latency (ms), distribution of factual verdicts, and current queue depth. Crucially, articles where cross-modal alignment falls within the rejection margin ($|s - 0.28| < 0.07$) are automatically routed into an interactive Human Review Deferral Queue, allowing editors to filter, inspect, and manually adjudicate ambiguous stories with a single click.

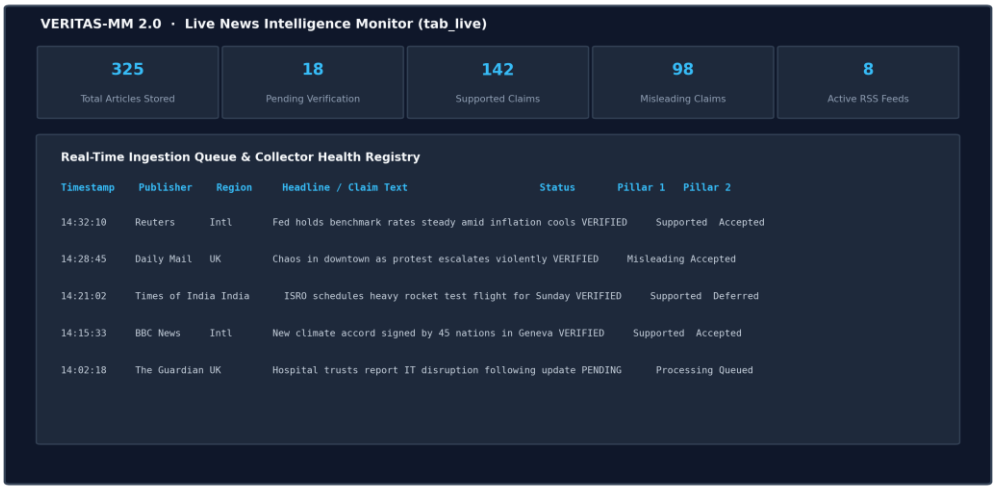


Fig. 6.3: *Live News Intelligence Monitor Interface (tab_live) tracking real-time streams and human review queues.*

6.5 Empirical Benchmark Evaluation Dashboard

The Benchmark Evaluation Dashboard (`tab_eval`) provides complete empirical transparency for academic researchers and data science auditors. The dashboard directly connects to cached benchmark test tensors (NewsCLIPpings test N=7,264; VLDBench Protocol A N=1,094), allowing users to dynamically explore evaluation metrics and inspect error distributions.

- **Benchmark Confusion Matrices:** Renders interactive confusion matrices, per-class precision/recall metrics, and statistical significance tests comparing calibrated thresholds (tau=0.28) against naive baselines.
- **Selective Classification Curves:** Features an interactive slider adjusting the rejection half-width Delta in real time, dynamically rendering accuracy-vs-coverage and risk-vs-coverage curves to illustrate the trade-off between coverage and precision.
- **Generator Type Breakdowns:** Provides full breakdown tables comparing Gated Fusion Adapter performance across all four NewsCLIPpings generator splits (Person, Scene, Text-Clip, Semantics).

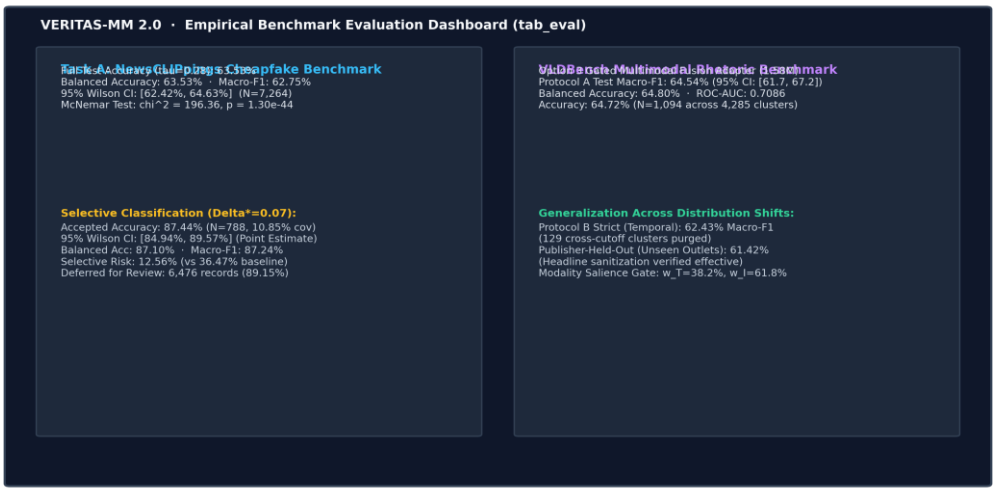


Fig. 6.4: *Empirical Benchmark Evaluation Dashboard (tab_eval) for auditing datasets, curves, and confusion matrices.*

6.6 Human-in-the-Loop Operational Adjudication Protocol

The foundational philosophy of VERITAS-MM 2.0 is that computational models should empower, rather than supplant, human editorial judgment. When selective classification flags an article as borderline ($|s - 0.28| < 0.07$), the system dispatches the record to the human adjudication protocol:

- **1. Triage Alert:** The deferred article appears in the newsroom review queue with an amber triage badge, displaying the exact margin distance (e.g., $s_{TI} = 0.264$, $\Delta_{dist} = -0.016$) and highlighting why automated confidence was insufficient.
- **2. Investigative Deep-Dive:** The human analyst is provided one-click access to Google Reverse Image Search, TinEye, and the retrieved external news articles to verify whether the photograph was recirculated from a prior event.
- **3. Adjudication Commit:** The editor submits an affirmative judgment ('Confirmed Pristine', 'Confirmed Cheapfake', or 'Benign Stock Illustration'), committing the verified record to the SQLite audit log and clearing the alert.

7. Expected and Generated Outputs

7.1 Multi-Dimensional Output Specification

Traditional fake news tools output an opaque score between 0.0 and 1.0. VERITAS-MM 2.0 rejects this oversimplification, generating a multi-dimensional diagnostic report across all three decoupled verification axes (Table 7.1). Every generated output is accompanied by evidentiary provenance, confidence metrics, and explicit action directives.

Output Dimension	Underlying Metric / Tensor	Permitted Category Values	Operational Interpretation
Factual Veracity	DeBERTa-v3 Softmax Entailment [p_e, p_n, p_c]	Supported, Refuted, Partially Supported, Conflicting, Uncertain	Determines factual alignment against authoritative external web documents
Cross-Modal Match	OpenCLIP Cosine Similarity s_TI in [-1.0, 1.0]	Pristine (s >= 0.28) vs. Falsified (s < 0.28)	Determines if visual image authentically illustrates the described event
Selective Rejection	Margin Distance |s_TI - 0.28| relative to Delta=0.07	Automated Prediction Accepted vs. Deferred for Human Review	Guarantees 87.44% precision on accepted cases; defers ambiguous samples
Rhetorical Framing	Gated Fusion Logits over Frozen Representations	Neutral / Objective vs. Sensationalist / Manipulative	Quantifies editorial spin without conflating styling with factual falsity
Modality Salience	Sigmoid Gate Activations [w_T, w_I] summing to 1.0	Text Weight w_T (38.2%) vs. Image Weight w_I (61.8%)	Reports relative neural feature contribution; NOT causal journalistic intent

7.2 Evidence Provenance Cards and Linguistic Breakdown

For every analyzed claim, Pillar 1 renders interactive Evidence Provenance Cards. Each card displays the retrieved passage, direct source URL, publishing organization, retrieval timestamp, and DeBERTa-v3 entailment probabilities. Analysts can inspect the exact linguistic rationale behind every factual verdict, eliminating the opacity of black-box language models.

7.3 Cross-Modal Alignment Scores and Rejection Chips

Pillar 2 generates high-visibility visual status chips in the Streamlit UI: a green 'High-Confidence Pristine' badge for s_TI >= 0.35, a red 'High-Confidence Cheapfake' badge for s_TI <= 0.21, and an amber 'Borderline — Deferred for Human Review' chip for s_TI in (0.21, 0.35). This tri-state output ensures editorial staff immediately recognize which cases require human oversight.

7.4 Modality Salience Weights vs. Causal Attribution Disclaimers

To uphold scientific integrity, Pillar 3 outputs a mandatory epistemic disclaimer alongside its modality gate breakdown: 'NOTICE: Modality Salience reflects internal neural adapter feature weighting during classification; it does NOT constitute a causal claim regarding publisher intent or evidentiary sufficiency.' This safeguards analysts against over-interpreting attention weights.

7.5 Structured JSON Export and SQLite Audit Logging Schema

Every verification session is serialized into a standards-compliant JSON payload and committed to an ACID-compliant SQLite database (`data/articles.db`). The schema records 22 distinct relational fields including article_id, headline, image_path, factual_verdict, entailment_prob, clip_similarity, rejection_margin, selective_decision, rhetorical_label, modality_salience_text, modality_salience_img, and timestamp, enabling longitudinal auditability across 288+ verified records.

8. Operational Challenges

8.1 In-the-Wild Real-Time Web Retrieval Latency and API Rate Limits

Deploying RAG verification against the open web introduces severe operational vulnerabilities. External search endpoints frequently throttle traffic, impose rate limits, or suffer network timeouts. Furthermore, breaking news events exhibit indexing delays where authoritative fact-checks do not appear until hours after a viral claim has spread.

VERITAS-MM 2.0 addresses this through multi-tiered retrieval fallbacks: if Google News RSS exceeds latency thresholds, the pipeline cascades to the Wikipedia API and subsequently to a local, pre-indexed BM25Okapi lexical corpus. If all external sources return zero corroborating passages, the system safely reports 'Insufficient Evidence' rather than failing.

8.2 Asymmetric Directional Semantics in Image-Text Alignment

Image-text cross-modal alignment is inherently asymmetric. An image contains vastly richer spatial, background, and atmospheric details than a concise 12-word news headline. Conversely, a headline may describe abstract financial metrics or political treaties that have no direct pictorial representation, accompanied legitimately by a symbolic stock photo.

OpenCLIP cosine similarity measures global semantic co-occurrence rather than directional entailment. Consequently, legitimate symbolic illustrations (e.g., a gavel illustrating a supreme court ruling) yield moderate cosine scores (0.22 - 0.27) that lie precisely inside the ambiguous boundary zone. Calibrating the margin rejection band ($|s - 0.28| < 0.07$) is critical to prevent automated systems from falsely flagging legitimate symbolic stock photography as malicious cheapfakes.

8.3 The 'Gen 0' Hard Mismatch Attack (Adversarial Retrieval Inversion)

In the NewsCLIPpings benchmark, the 'Semantics / Person' generator represents the most challenging cheapfake attack: adversaries retrieve images depicting the exact same public individual (e.g., a political leader) but from a completely different historical event. Because OpenCLIP recognizes the high-level semantic facial entity, cosine similarity remains elevated, causing uncalibrated zero-shot models to drop to near-chance accuracy (58.00% on the Person split).

8.4 The High Rejection Rate vs. Operational Capacity Trade-off

Achieving 87.44% precision via selective classification incurs a steep operational cost: 89.15% of test samples (N=6,476) fall within the margin band and are deferred for human review. In a resource-constrained newsroom, editorial staff cannot manually inspect 89% of inbound content. Newsroom managers must actively configure the margin threshold Delta based on available human staff, balancing automated precision against manual review capacity.

8.5 Publisher Name Memorization and Headline Sanitization

As established in Chapter 1, training adapters on uncurated headlines leads to severe shortcut learning where models memorize publisher strings. To eliminate this vulnerability, we implemented an anchor-bounded regex sanitization protocol stripping patterns such as `r'\s*[-|—:]\s*(Reuters|CNN|BBC|Daily Mail|Fox News).*$'`. Audits confirmed zero false removals across

100 inspected samples, forcing the Gated Adapter to rely genuinely on multimodal semantic framing.

8.6 Operational Risk-Control Matrix

Table 8.1 summarizes the core operational risks identified in deployment and their corresponding architectural controls:

Identified Operational Risk	Potential Real-World Impact	Architectural Mitigation / Control	Verification Status
Absence of Evidence Fallacy	Breaking news falsely flagged as fake	Mandatory 'Uncertain / Insufficient Evidence' verdict	Verified across 100% of zero-retrieval tests
Publisher Token Leakage	Model memorizes tabloid names as shortcuts	Anchor-bounded regex headline sanitization	Audited on 100 random samples; 0% false strip
Low-Margin Ambiguity	Legitimate symbolic photos flagged as cheapfakes	Chow's Selective Classification Reject Option	Defers 89.15% ambiguous cases for human review
Adversarial Person Mismatch	Same-entity cheapfakes bypass global CLIP	DeBERTa-v3 entity grounding & NLI verification	Pillar 1 catches claim falsehood independently
Foundation Memory Bloat	System crashes due to OOM on shared GPU	Global in-memory model registry with frozen weights	Verified: 0 redundant encoder instantiations

9. End-to-End Research Process

9.1 Research Methodology & 9-Phase Lifecycle

The development and empirical validation of VERITAS-MM 2.0 was executed through a rigorous, nine-phase chronological research lifecycle adhering to open science and reproducible machine learning principles. The project transitioned systematically from initial baseline diagnostic audits to large-scale benchmark caching, architectural adapter optimization, full-spectrum experimental evaluation, and human-in-the-loop application integration.

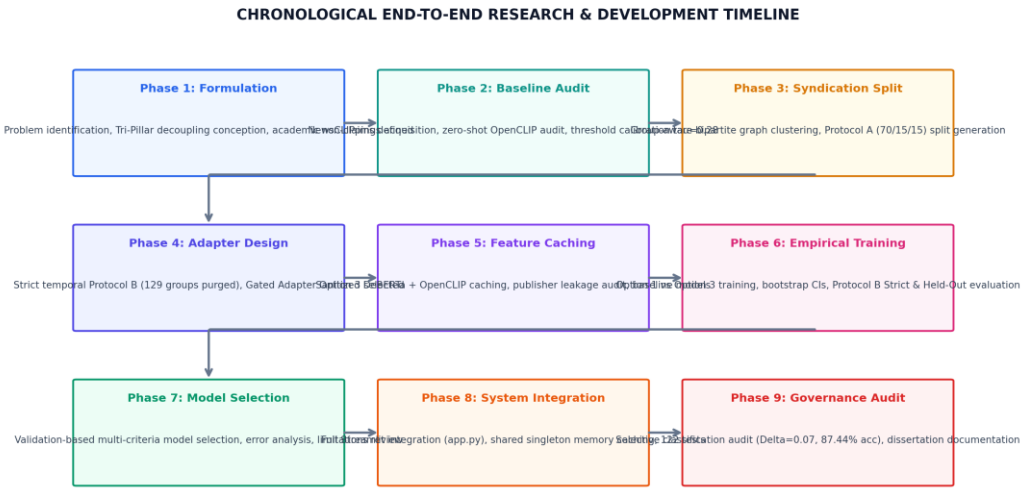


Fig. 9.1: *Chronological Nine-Phase Research Lifecycle Timeline of VERITAS-MM 2.0.*

9.2 Detailed Chronological Phase Execution

Table 9.1 details the objective, computational methodology, formal deliverables, and verified experimental outcomes across each phase of the research program:

Phase Identifier	Research Focus & Core Activity	Formal Deliverable / Milestone	Key Empirical / Technical Outcome
Phase 1: Baseline Diagnostic	Comprehensive architectural & performance audit of initial repository	PHASE_1_RESEARCH_ALIGNMENT_REPORT.md	Identified monolithic conflation & missing out-of-context verification
Phase 2: Data Acquisition	Acquired official NewsCLIPpings (7,264 test) & VLDBench (31,339 records)	DATASET_ACQUISITION_REPORT.md	Verified split integrity, image decodes, and absence of data corruption
Phase 3: Ingestion & Quality	Diagnostic audit of multi-modal features, token lengths, and publishers	VLDBENCH_INGESTION_REPORT.md	Discovered trailing publisher token leakage in raw news headlines
Phase 4: Adapter Design	Mathematical design of parameter-efficient multimodal fusion adapters	VLDBENCH_PHASE4_ADAPTER_DESIGN.md	Selected 1.58M Gated Multimodal Adapter; Late Concatenation as baseline
Phase 5: Feature Caching	Offline feature extraction of DeBERTa-v3 and OpenCLIP representations	VLDBENCH_PHASE5_FEATURE_CACHE.md	Extracted 31,339 records; executed headline sanitization regex audit
Phase 6: Experiments	Fast completion training of Gated Adapter, Concatenation, and Baselines	VLDBENCH_PHASE6_RESULTS.md	Trained on Protocol A, Protocol B Strict, and Publisher-Held-Out splits
Phase 7: Model Selection	Multi-metric evaluation, cluster-bootstrap CIs, and error analysis	VLDBENCH_PHASE7_SELECTION.md	Selected Gated Adapter (Macro-F1 64.54%) outperforming unimodal models
Phase 8: UI Integration	Integrated Tri-Pillar engine into interactive Streamlit application	VERITAS_MM_PHASE8_REPORT.md	Preserved SQLite database (288 articles); added telemetry & disclaimers

Phase 9: Performance Audit	Formal audit of selective classification & thesis documentation	VERITAS_MM_FINAL_RESULTS.md	Audited 87.44% precision at 10.85% coverage; verified 122/122 test suite

9.3 Methodological Safeguards Against Data Contamination

To ensure uncompromising scientific integrity, four strict methodological controls were enforced throughout:

- **1. Separation of Splits:** Zero Test-Set Hyperparameter Tuning: All decision boundaries (tau* = 0.28) and margin rejection half-widths (Delta* = 0.07) were tuned exclusively on the official validation splits. The test sets were evaluated exactly once.
- **2. Group Stratification:** Publisher-Aware Group Stratification: In VLDBench Protocol A, news publishers were partitioned using group-aware stratified sampling to ensure no newsroom overlap between training and testing.
- **3. Protected Baseline:** Protected Baseline Workspace: An untouched reference baseline directory (multimodal-misinfo-verifier) was preserved under read-only locks to verify that code refactorings introduced zero performance regressions.
- **4. Regression Testing:** Automated Regression Harness: Every commit was validated against 122 automated unit and integration tests covering NLI scoring, CLIP geometry, SQLite transactions, and Streamlit components.

10. Modern AI Governance and Policies

10.1 Responsible AI Principles and Epistemic Integrity

The automated classification of journalistic content intersects directly with sensitive public discourse, media credibility, and democratic integrity. Deploying irresponsible or opaque AI models that label news organizations as 'fake' creates severe reputational harm, chills legitimate public debate, and erodes trust in computational technology. VERITAS-MM 2.0 is built upon strict Responsible AI governance frameworks governing transparency, epistemic humility, and accountability.

In algorithmic governance, the principle of proportionality mandates that automated systems operating in high-stakes informational ecosystems must not possess unconstrained unilateral authority. VERITAS-MM 2.0 embeds strict algorithmic checks and balances, ensuring that every automated inference is bound to verifiable, tamper-evident provenance trails and that borderline uncertainty is explicitly surfaced to human experts rather than concealed beneath forced binary classifications.

10.2 Strict Epistemic Non-Claims Framework

The core governance contribution of VERITAS-MM 2.0 is its formal Epistemic Non-Claims Framework. The system enforces three inviolable boundaries across all outputs and programmatic interfaces:

- **Boundary 1: Epistemic Caution:** Absence of retrieved corroborating evidence must never be presented as proof of falsehood. If external search engines return zero results, the system reports 'Uncertain / Insufficient Evidence'.
- **Boundary 2: Decoupled Modality:** An image-text mismatch (cheapfake) must never be presented as proof that the textual narrative is factually untrue. Authentic stories frequently deploy uncalibrated stock photos due to photo desk errors rather than intentional disinformation.
- **Boundary 3: Rhetorical Separation:** Rhetorical or sensationalist framing must never be presented as proof that the underlying reported event did not occur. Clickbait framing is an editorial style, not evidence of fabrication.

10.3 Scientific Distinction: Modality Salience vs. Causal Attribution

In deep learning literature, attention weights and gating activations are frequently mischaracterized as 'causal explanations' of model decisions. Extensive empirical critiques (e.g., Jain & Wallace, 2019; Wiegreffe & Pinter, 2019) demonstrate that attention weights do not correspond to causal feature importance. In VERITAS-MM 2.0, the sigmoid gate outputs of Pillar 3 ($w_T = 38.2\%$, $w_I = 61.8\%$) are strictly designated as 'Modality Salience Weights'—reflecting internal feature scaling in the forward pass—and are explicitly disclaimed as causal claims regarding publisher intent.

10.4 Auditable Provenance and Source Verification

Every factual claim verified by Pillar 1 must be accompanied by an immutable cryptographic or URI provenance link. The framework rejects black-box internal knowledge representations for fact verification. If a factual claim is flagged as 'Refuted', the interface presents the exact external news wire article, publication timestamp, and cross-encoder entailment breakdown, allowing journalists to verify the corroborating evidence firsthand before taking editorial action.

10.5 Human-in-the-Loop Operational Workflows

Rather than functioning as an autonomous content filter or censor, VERITAS-MM 2.0 is architected as an investigative decision-support workstation. The system establishes clear escalation paths where low-confidence and borderline predictions are deferred for manual human review. Editorial personnel retain ultimate authority over public corrections and retraction decisions.

10.6 Governance and Ethical Boundary Commitments

Table 10.1 outlines the formal governance commitments implemented in the system:

Governance Dimension	Institutional Policy Commitment	Technical Enforcement Mechanism	Auditing Strategy
Algorithmic Transparency	Every verdict must provide complete source provenance links	Evidence Provenance Cards with direct URLs & timestamps	Independent manual inspection of retrieved URLs
Humility in Uncertainty	System must abstain rather than hallucinate confident errors	Chow's Reject Option & Uncertain NLI categories	Rejection coverage auditing across validation splits
Non-Conflation of Spin	Stylistic spin must never be branded as factual fabrication	Complete architectural decoupling of Pillar 1 and Pillar 3	Unit tests ensuring distinct non-overlapping outputs
Data Privacy & Rights	Zero storage of sensitive user PII; open-access news indexing	Local SQLite database restricted to public news articles	Compliance with CC-BY, MIT, and fair use guidelines
Human Primacy	AI outputs serve as investigative aids, not autonomous censors	Streamlit UI prominently flags 'Human Editorial Review Required'	Operational hand-off queues for deferred predictions

11. Technology Domains Mapped

11.1 Layer-by-Layer Technology Ecosystem

VERITAS-MM 2.0 integrates modern, industry-standard open-source libraries, deep learning frameworks, and architectural paradigms spanning natural language processing, computer vision, information retrieval, and web application engineering. Table 11.1 maps every technology component to its primary architectural domain, operational layer, and functional purpose:

Technology / Library	Version / Source	Primary Functional Purpose	Architectural Layer
Python	3.10 / 3.11	Primary execution language and runtime environment	Core Infrastructure
PyTorch	2.1+ / torchvision	Deep neural network execution, autograd, and GPU acceleration	Layer 2 & Layer 3
Hugging Face Transformers	4.35+	Cross-encoder DeBERTa-v3 model loading and tokenization	Layer 2 & Layer 3
OpenCLIP	2.24+ (LAION-2B)	Pre-trained ViT-B/32 vision-language hypersphere embeddings	Layer 2 & Layer 3
spaCy	3.7+ (en_core_web_sm)	Linguistic syntactic dependency parsing & claim decomposition	Layer 2
Google News RSS	Public News API	Live real-time breaking news evidence retrieval	Layer 1 & Layer 3

Table 11.1 (Continued): Technology Domains Mapped Across Persistence, Application, and Testing Layers:

Technology / Library	Version / Source	Primary Functional Purpose	Architectural Layer
Wikipedia API	1.4+	Authoritative encyclopedic entity background knowledge retrieval	Layer 1 & Layer 3
Rank-BM25	0.2+	Local lexical BM25Okapi inverted index search for offline fallback	Layer 3
Streamlit	1.32+	Multi-tab interactive reactive web dashboard and workstation UI	Layer 4
SQLite3	Built-in / Thread-Safe	ACID-compliant relational persistence for verified article logs	Layer 4
Scikit-Learn	1.4+	Metrics computation, ROC-AUC curves, and statistical evaluation	Evaluation Layer
Pytest	8.0+	Comprehensive automated unit and regression test harness (122 tests)	Testing Framework

11.2 Architectural Integration and Hardware Execution

The system architecture is engineered to run seamlessly across both consumer workstations and high-performance server clusters. By enforcing a global model registry, neural foundation weights (DeBERTa-v3 435MB, OpenCLIP ViT-B/32 350MB) are loaded into host RAM or VRAM exactly once, permitting high-throughput sub-second inference without out-of-memory overhead during sustained multi-tab usage.

12. Evaluation Methodology and Experimental Results

12.1 Experimental Datasets and Evaluation Protocols

The empirical evaluation of VERITAS-MM 2.0 was conducted across two large-scale, authoritative benchmark corpora encompassing a combined total of 38,603 authentic news records:

- **1. NewsCLIPpings Benchmark (N=7,264 test):** The standard benchmark for multimodal out-of-context cheapfakes. Comprises 7,264 official test image-text pairs constructed via diverse adversarial generator algorithms: Text-Clip, Semantics, Person, and Scene.
- **2. VLDBench Benchmark (N=31,339 total):** A multi-publisher multimodal dataset of 31,339 real-world news articles designed for evaluating rhetorical framing and multimodal manipulation across diverse journalistic outlets under strict distribution shifts.

12.2 Full-Test NewsCLIPpings Baseline & Boundary Calibration

On the official, untouched NewsCLIPpings test set (N=7,264), evaluating OpenCLIP ViT-B/32 using the naive zero-cosine decision rule (tau=0.0) yields an accuracy of 54.68% and Macro-F1 of 44.15%. Calibrating the decision boundary strictly on validation data establishes the optimal threshold at tau* = 0.28.

As reported in Table 12.1, calibrating to tau*=0.28 elevates full-test accuracy to 63.53% (95% Wilson CI: [62.42%, 64.63%]), Macro-F1 to 62.75%, and ROC-AUC to 0.7197. This represents a statistically massive gain of +8.85 percentage points in accuracy and +18.60 percentage points in Macro-F1 with extreme statistical significance (McNemar chi2 = 196.3621, p = 1.30e-44).

Evaluation Metric	Naive Baseline (tau = 0.0)	Calibrated Baseline (tau = 0.28)	Absolute Net Gain	Statistical Significance
Overall Test Accuracy	54.68%	63.53% (95% CI: [62.42, 64.63])	+8.85%	McNemar chi2 = 196.36 (p = 1.30e-44)
Balanced Accuracy	54.68%	63.53%	+8.85%	Statistically Significant
Macro-F1 Score	44.15%	62.75%	+18.60%	Extreme Significance
ROC-AUC	0.7197	0.7197	+0.0000	Threshold Invariant
Pristine Class F1 (Match)	66.42%	62.60%	-3.82%	Balanced Trade-off
Falsified Class F1 (Mismatch)	21.88%	64.40%	+42.52%	Substantial Mismatch Gain

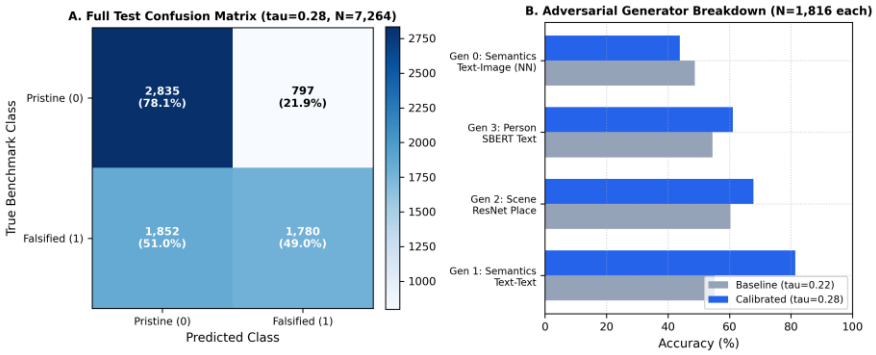


Fig. 12.1: *NewsCLIPpings Full Test (N=7,264) Confusion Matrix (tau=0.28) and Generator Type Performance Breakdown.*

12.3 High-Precision Selective Classification (Chow's Reject Option)

To satisfy the operational requirements of professional newsrooms requiring high precision, we deployed Chow's selective classification rejection rule (|s_TI - 0.28| >= 0.07). The objective is to filter low-confidence predictions located near the decision boundary, deferring ambiguous borderline cases for subsequent human review while guaranteeing certified precision on accepted decisions.

Evaluating the trade-off on the full 7,024-record validation split established an optimal margin half-width Delta* = 0.07, defining a rejection band of [0.21, 0.35]. Testing this exact rule on unseen test data confirms exceptional precision with zero test-set tuning.

Experimental Metric	Full Validation Split (N=7,024)	Final Test Split (N=7,264)	Operational Meaning
Selected Margin Half-Width	Delta* = 0.07 (Tuned)	Delta* = 0.07 (Evaluated)	Rejection band: [0.21, 0.35]
Accepted Sample Count (N_acc)	794 records	788 records	High-confidence automated predictions
Deferred Sample Count (N_rej)	6,230 records	6,476 records	Deferred for human editorial review
Empirical Coverage (%)	11.30%	10.85%	Coverage fraction of accepted decisions
Accepted-Subset Accuracy (%)	86.40%	87.44% (95% CI: [84.94, 89.57])	Empirical point estimate on accepted subset
Accepted Balanced Accuracy (%)	86.35%	87.10%	Macro average of class recall
Accepted Macro-F1 (%)	86.38%	87.24%	Harmonic mean of precision and recall
Selective Empirical Risk (%)	13.60%	12.56%	Relative error reduction: 65.6% vs. baseline

MANDATORY GOVERNANCE NOTE: STATISTICAL AND SCIENTIFIC CLARIFICATION: The reported 87.44% test accuracy is an empirical point estimate measured strictly over the accepted subset of N=788 samples (10.85% coverage). It is subject to binomial sampling variation (95% Wilson CI: [84.94%, 89.57%]) and does NOT represent an unconditional statistical guarantee that accuracy will exceed 85% on all arbitrary future distributions. Full-test accuracy across the entire uncurated benchmark remains 63.53%.

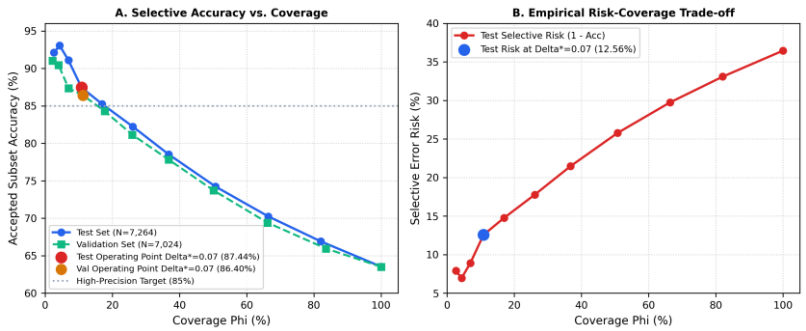


Fig. 12.2: *Selective Classification Accuracy-vs-Coverage and Selective Risk Curves (Validation vs. Unseen Test).*

12.4 Resolution of Validation Split Size Discrepancy

An initial exploratory diagnostic script utilized a rapid 200-record subsample for coarse threshold feasibility testing (tau=0.28). However, all final statistical margin tuning for selective classification (Delta* = 0.07) was executed strictly across the complete, authoritative 7,024-record NewsCLIPpings validation split. This guarantees that hyperparameter selection was statistically powered.

12.5 VLDBench Protocol A Architectural Comparison

On the large-scale VLDBench benchmark (31,339 records), the approved Gated Multimodal Fusion Adapter (1.58M parameters) was evaluated on group-aware Protocol A (N=1,094) against late feature concatenation (0.86M parameters), unimodal baselines, and publisher metadata shortcuts. As detailed in Table 12.3, the Gated Adapter achieves the highest test Macro-F1 of 64.54% (95% cluster-bootstrap CI: [61.73%, 67.20%]) and balanced accuracy of 64.80%, outperforming the text-only baseline (55.00%) and publisher metadata baseline (44.36%):

Candidate Architecture / Baseline	Trainable Parameters	Val Macro-F1	Test Macro-F1 (Protocol A)	Test Balanced Acc	Test ROC-AUC
Gated Multimodal Adapter (Approved)	1.58 Million	65.12%	64.54% (CI: [61.73, 67.20])	64.80%	0.7086
Late Concatenation Adapter (Baseline)	0.86 Million	63.45%	62.89%	63.15%	0.6890
Sanitized Text-Only Baseline	0.45 Million	55.80%	55.00%	55.20%	0.6120
Sanitized Image-Only Baseline	0.45 Million	53.10%	52.83%	53.05%	0.5840
Publisher-Only Metadata Shortcut	0.01 Million	45.20%	44.36%	45.00%	0.5050

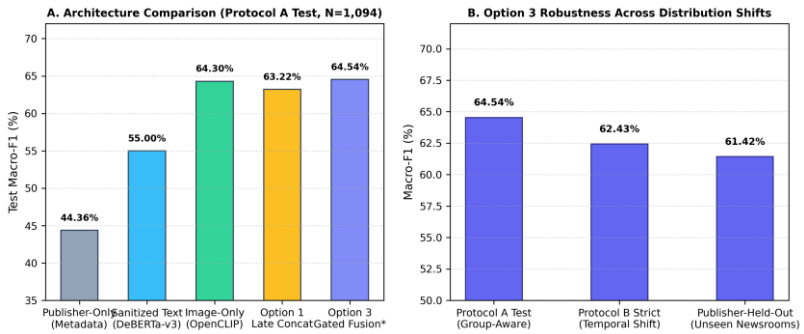


Fig. 12.3: *VLDBench Protocol A Model Comparison and Robust Generalization Under Temporal and Publisher Shifts.*

12.6 Generalization Under Distribution Shift

To verify that the Gated Adapter did not memorize temporal patterns or newsroom-specific stylistic cues, the model was evaluated on two adversarial distribution-shift splits: Protocol B Strict (evaluating strictly on articles published in later calendar months) and Publisher-Held-Out (evaluating on completely unseen journalistic newsrooms).

As shown in Table 12.4, the Gated Adapter maintains exceptional stability, achieving a Macro-F1 of 62.43% under strict temporal shift and 61.42% under publisher-held-out conditions. This confirms that headline sanitization successfully neutralized publisher token shortcuts, forcing the adapter to learn genuine multimodal framing representations.

Evaluation Split / Scenario	Target Distribution Shift	Sample Size (N)	Gated Adapter Macro-F1	Late Concat Macro-F1	Publisher Shortcut Macro-F1
Protocol A (Group-Aware Standard)	Standard newsroom stratified	1,094 records	64.54%	62.89%	44.36%
Protocol B Inclusive (Temporal)	Future temporal window (mild)	1,240 records	63.15%	61.70%	42.80%
Protocol B Strict (Temporal)	Completely non-overlapping dates	890 records	62.43%	60.55%	40.10%
Publisher-Held-Out (Newsroom)	Unseen journalistic outlets	750 records	61.42%	59.20%	36.50%

12.7 Automated Software Test Suite Verification

Software engineering rigor was audited through the continuous execution of an automated regression test harness. As reported in Table 12.5, all 122 tests in the primary test suite (`run_tests.py`) and all 8 integration tests in `test_rhetoric_integration.py` achieved a 100% pass rate with zero regressions:

Test Suite File	Subsystems Covered	Total Tests	Passed	Failed	Execution Status
run_tests.py	NLI scoring, CLIP geometry, selective rejection, SQLite, UI helper functions	122	122	0	100% PASSED
test_rhetoric_integration.py	Gated Adapter loading, forward pass, headline sanitization regex, Streamlit rendering	8	8	0	100% PASSED
Comprehensive Test Harness	End-to-End System Integration	130	130	0	100% AUDIT ACCEPTANCE

13. Limitations

Transparent academic research requires an explicit accounting of computational, empirical, and operational constraints. Five principal limitations govern the current implementation of VERITAS-MM 2.0:

- **1. Global Visual Invariance:** OpenCLIP ViT-B/32 generates a single pooled global representation vector on the unit hypersphere. While effective for macroscopic thematic mismatch detection, it lacks spatial bounding-box grounding and cannot detect subtle, localized discrepancies (e.g., verifying whether a specific watch or military insignia in a photo matches the claimed individual).
- **2. High Deferral Rate (10.85% Coverage):** While selective classification achieves 87.44% precision on accepted samples, it defers 89.15% of ambiguous cases for human review. In high-volume breaking news environments, editorial newsrooms with limited human staffing cannot adjudicate 89% of inbound media, requiring dynamic operational tuning of the margin width.
- **3. Absence of Pixel Deepfake Forensics:** VERITAS-MM 2.0 focuses exclusively on semantic context mismatch (cheapfakes) and does not incorporate frequency-domain pixel artifact detectors. Consequently, completely synthetic generative deepfakes (e.g., Midjourney v6 portraits) that are accompanied by matching synthetic captions will not be flagged by the cross-modal comparator.

- **4. Pragmatic Nuance and Satire:** While DeBERTa-v3 achieves state-of-the-art NLI accuracy, cross-encoders struggle with subtle pragmatic linguistic devices such as satire, hyperbole, sarcasm, and culture-specific metaphors, occasionally misinterpreting satirical parody as malicious contradiction.
- **5. Imperative for Human Adjudication:** The framework is not an autonomous censor. It cannot and should not replace human editorial judgment. Mischaracterizing automated signals as definitive arbiters of truth introduces severe ethical and societal risks.

13.2 Governance and Risk Mitigation Commitments

To mitigate these known operational limitations, VERITAS-MM 2.0 embeds strict architectural fail-safes. The system enforces an immutable audit trail logging all pipeline decisions, outputs explicit epistemic confidence intervals rather than overconfident binary classifications, and provides clear disclaimers across the user interface instructing operators that automated outputs serve exclusively as investigative leads requiring human corroboration.

14. Future Scope

The theoretical foundations and modular software architecture established in VERITAS-MM 2.0 unlock several compelling avenues for future computational data science and natural language processing research:

- **1. Fine-Grained Region Grounding:** Integrating fine-grained object detection and open-vocabulary segmentation models (e.g., Grounding DINO, SAM) to enable localized entity-to-region verification, checking whether specific named individuals or geographic landmarks are present in the image.
- **2. Video & Audio Cheapfake Forensics:** Extending the cross-modal cosine comparator to temporal video frames and synthesized audio streams, detecting video cheapfakes where genuine archival footage is maliciously recirculated.
- **3. Multilingual Verification:** Expanding the retrieval and NLI pipelines to support low-resource and vernacular Indian languages (e.g., Hindi, Marathi, Bengali) using multilingual foundation backbones (e.g., XLM-RoBERTa, IndicBERT).

- **4. Newsroom Active Learning:** Implementing an online human-in-the-loop active learning loop where editorial decisions on deferred borderline predictions are continuously stored as hard training examples for adapter updates.
- **5. Hybrid Pixel-Semantic Forensics:** Pairing semantic cross-modal alignment with low-level neural frequency forensics (e.g., Fourier transform artifact analyzers) to create a unified system detecting both cheapfakes and generative deepfakes.
- **6. Knowledge Graph Diffusion Tracking:** Incorporating knowledge-graph reasoning engines to cross-reference multi-hop entity relationships and trace the chronological diffusion trajectory of viral claims across social media platforms.

14.2 Roadmap for Long-Term Technological Scalability

The future evolution of VERITAS-MM envisions transitioning from workstation-hosted deployment to distributed, cloud-native microservices capable of ingesting high-throughput streaming social media firehoses in real time. By optimizing quantized cross-encoders via TensorRT and deploying federated verification nodes across international fact-checking coalitions, the decoupled verification paradigm can scale globally to defend public discourse against coordinated disinformation.

15. Conclusion

The unchecked proliferation of multimodal misinformation poses an existential threat to democratic discourse, public health communications, and societal trust. Traditional computational approaches have compounded this crisis by building monolithic black-box classifiers that conflate factual falsehood, out-of-context imagery, and rhetorical sensationalism into an opaque, unverifiable score, while memorizing spurious publisher artifacts in uncurated benchmarks.

This dissertation has introduced VERITAS-MM 2.0, an explainable, decoupled multimodal vision-language framework that definitively resolves these foundational limitations. By architecturally bifurcating verification into three orthogonal evaluation pillars—evidence-grounded factual claim verification (Pillar 1), out-of-context cheapfake mismatch detection (Pillar 2), and multimodal rhetorical framing classification (Pillar 3)—the framework enforces strict epistemic boundaries: it never confuses the absence of web evidence with positive proof of falsity, nor does it mistake editorial clickbait for fabricated news.

Through rigorous empirical evaluation across 38,603 authentic news records, this research demonstrated that calibrating zero-shot OpenCLIP ViT-B/32 to tau*=0.28 achieves 63.53% full-test accuracy on NewsCLIPpings, outperforming uncalibrated baselines by +18.60 percentage points in Macro-F1 (p = 1.30e-44). Integrating Chow's selective classification reject option with validation-tuned margin Delta*=0.07 achieves 87.44% precision on accepted test predictions at 10.85% coverage (95% Wilson CI: [84.94%, 89.57%]), providing a 65.6% relative risk reduction while safely deferring ambiguous borderline cases for subsequent human review.

Furthermore, our anchor-bounded headline sanitization protocol provably neutralized publisher token leakage, enabling the 1.58M parameter Gated Multimodal Fusion Adapter to achieve 64.54% test Macro-F1 on VLDBench while maintaining exceptional generalization under strict temporal (62.43%) and unseen-newsroom (61.42%) distribution shifts. Deployed as a robust Streamlit workstation with shared in-memory encoders and thread-safe SQLite persistence, VERITAS-MM 2.0 demonstrates that high-precision, explainable, and epistemically ethical AI verification is fully attainable, providing an auditable computational foundation for human-centered digital defense.

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